DNS: How a Website Name Finds Its Address

Explain it in your own words

Use this guide to understand each animation, choose your phrasing, and keep a natural YouTube delivery. Every clip has a matching frame, the idea behind the movement, and a specific delivery cue.

16 clips | 09:29.6 continuous video | 10 guide pages

A simple speaking rhythm

1. Set up: Tell viewers what the objects represent and ask one small question.

2. Follow the change: Point to what moves, grows, disappears or gets replaced.

3. Explain the cause: Say why that change happened. Use the mechanism notes as your checklist.

4. Let it land: Pause on the result, then connect it to the next question.

Use the timeline flexibly

The next page gives exact clip boundaries in the current continuous MP4. Individual clip files begin at 00:00. In the self-paced player, a phase holds until you advance, so your own recording can be longer or shorter. These timestamps describe the video; delivery cues describe how you might explain it.

Press Right / Left to move between phases, Up / Down to adjust speed, R to replay, and Shift+P for the separate script window. Rehearse the cause-and-effect in your own words before recording.

Open this video in the local player | Full narration reference | Sources

The complete timeline

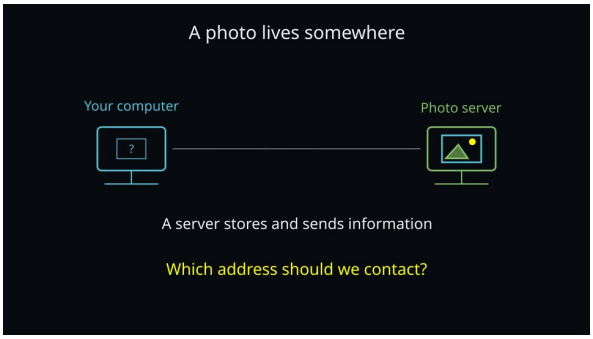
Find the clip, see the question being explained, and jump to its guide page. Times are MM:SS.t in the full MP4.

CLIP / TIME	WHAT YOU ARE EXPLAINING	PAGE
01 00:00.0 - 00:35.7	A photo lives somewhere	3
02 00:35.7 - 01:10.4	Names are easier to remember	3
03 01:10.4 - 01:45.6	Ask a helper	4
04 01:45.6 - 02:20.4	A name has smaller parts	4
05 02:20.4 - 02:56.5	Start at the directory	5
06 02:56.5 - 03:31.7	Follow one useful clue	5
07 03:31.7 - 04:06.9	Find the address keeper	6
08 04:06.9 - 04:42.6	Bring the answer home	6
09 04:42.6 - 05:17.8	Put a clock on it	7
10 05:17.8 - 05:53.4	Remember the answer	7
11 05:53.4 - 06:29.1	Memory needs an expiry date	8
12 06:29.1 - 07:04.8	What if the address changes?	8
13 07:04.8 - 07:41.3	Look again when time runs out	9
14 07:41.3 - 08:17.8	Connected, but missing directions	9
15 08:17.8 - 08:53.9	Three jobs, one simple search	10
16 08:53.9 - 09:29.6	From a name to an address	10

Use the clip number as your anchor when recording. Matching frames show the completed diagram; watch the movement too, because the change is often the explanation.

CLIP 01 / A photo lives somewhere

00:00.0 - 00:35.7 in full video | 35.7s clip



What the viewer sees
Two computers and a photo, joined by a connection.

What you need to explain - and why
A server stores information and sends it when asked. Before requesting the photo, the other computer needs a destination address.

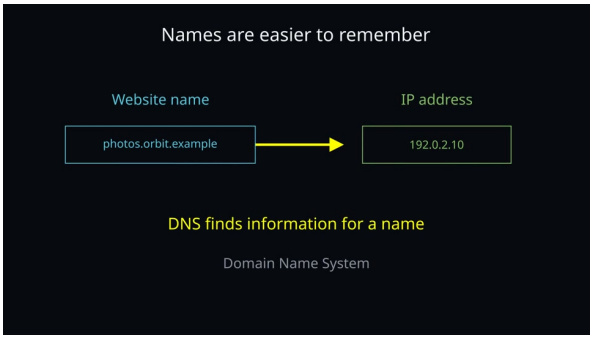
Tone and emphasis
Curious opening: where does a website name actually lead?

Next: Names are easier to remember

Open clip 01 | My phrasing: _____

CLIP 02 / Names are easier to remember

00:35.7 - 01:10.4 in full video | 34.8s clip



What the viewer sees
A familiar contact name becomes a website name and an IP address.

What you need to explain - and why
People remember names; the network uses addresses for delivery. DNS finds information associated with a name.

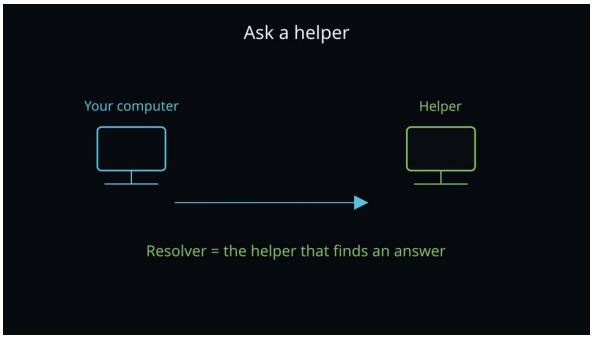
Tone and emphasis
Use the friend-and-phone-number comparison, then name DNS once.

Next: Ask a helper

Open clip 02 | My phrasing: _____

CLIP 03 / Ask a helper

01:10.4 - 01:45.6 in full video | 35.2s clip



What the viewer sees
A question travels from your computer to a green helper.

What you need to explain - and why
A resolver searches on your behalf and returns an answer. Begin with no saved answer so the search is visible.

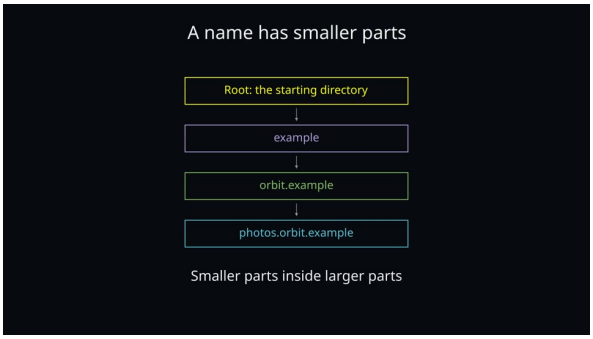
Tone and emphasis
Introduce the helper's job before saying resolver.

Next: A name has smaller parts

Open clip 03 | My phrasing: _____

CLIP 04 / A name has smaller parts

01:45.6 - 02:20.4 in full video | 34.8s clip



What the viewer sees
The name becomes a chain of smaller branches.

What you need to explain - and why
The rightmost parts identify broader responsibility. Each smaller branch narrows where the helper should ask.

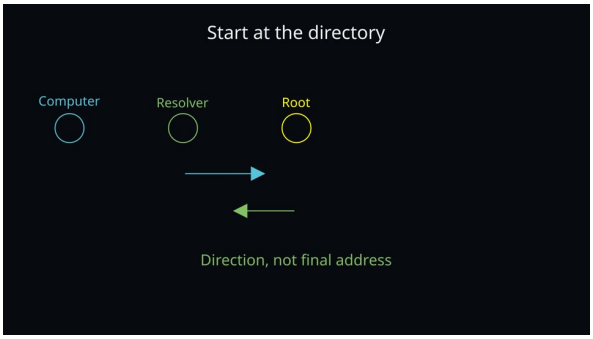
Tone and emphasis
Point from the broad directory toward the complete name.

Next: Start at the directory

Open clip 04 | My phrasing: _____

CLIP 05 / Start at the directory

02:20.4 - 02:56.5 in full video | 36.1s clip



What the viewer sees
The helper asks the root and receives a direction.

What you need to explain - and why
The starting directory can direct a search without knowing the final address. That direction is a referral.

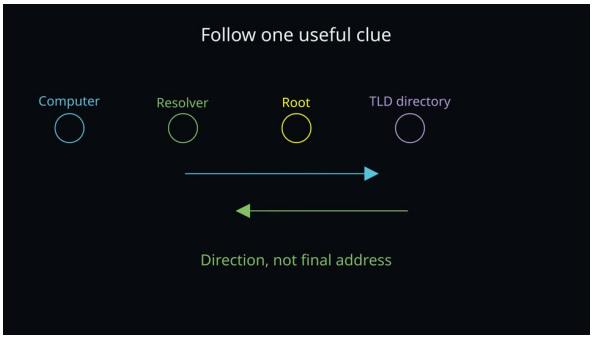
Tone and emphasis
Ask: can directions be useful even without the final answer?

Next: Follow one useful clue

Open clip 05 | My phrasing: _____

CLIP 06 / Follow one useful clue

02:56.5 - 03:31.7 in full video | 35.2s clip



What the viewer sees
Another directory appears and sends a narrower direction.

What you need to explain - and why
The top-level domain directory points toward the servers responsible for the next branch. The resolver keeps doing the asking.

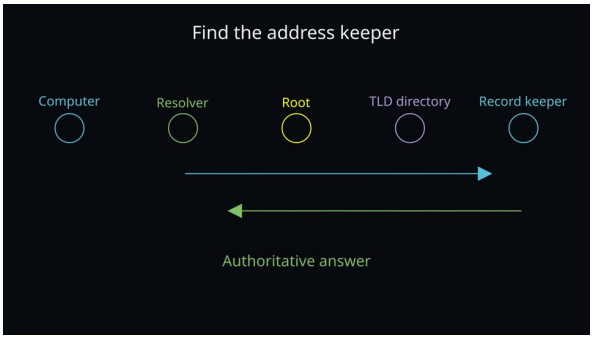
Tone and emphasis
Repeat the ask, get a clue, ask again pattern.

Next: Find the address keeper

Open clip 06 | My phrasing: _____

CLIP 07 / Find the address keeper

03:31.7 - 04:06.9 in full video | 35.2s clip



What the viewer sees
The record keeper returns a green address.

What you need to explain - and why
An authoritative server holds the relevant official record. Earlier directories guided the helper to this answer.

Tone and emphasis
Let the appearance of the actual address feel like a small payoff.

Next: Bring the answer home

Open clip 07 | My phrasing: _____

CLIP 08 / Bring the answer home

04:06.9 - 04:42.6 in full video | 35.7s clip



What the viewer sees
The answer reaches your computer.

What you need to explain - and why
The lookup supplies contact information. Fetching the photo is the next job, just as finding a number comes before a call.

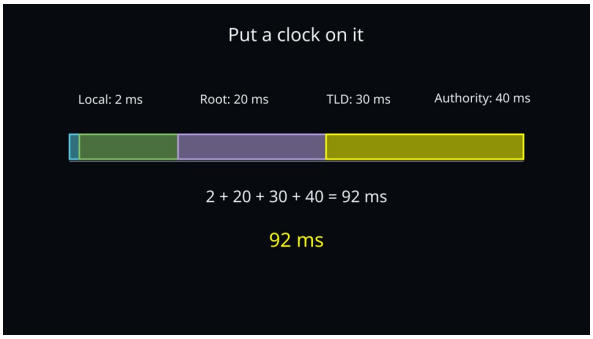
Tone and emphasis
Pause to separate finding the address from fetching the content.

Next: Put a clock on it

Open clip 08 | My phrasing: _____

CLIP 09 / Put a clock on it

04:42.6 - 05:17.8 in full video | 35.2s clip



What the viewer sees
Four colored waits grow into one time bar.

What you need to explain - and why
The example waits depend on previous answers, so add them: 2 + 20 + 30 + 40 = 92 milliseconds. A millisecond is one thousandth of a second.

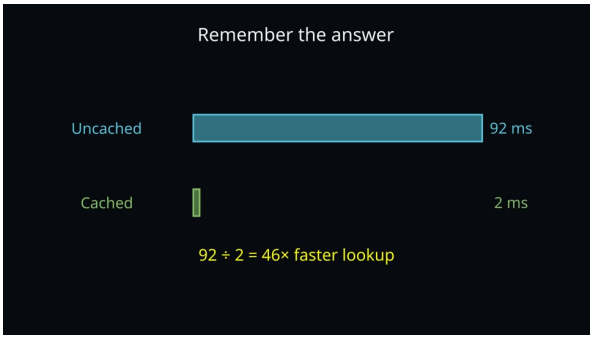
Tone and emphasis
Read the moving bar first, then the sum; these are chosen example timings.

Next: Remember the answer

Open clip 09 | My phrasing: _____

CLIP 10 / Remember the answer

05:17.8 - 05:53.4 in full video | 35.7s clip



What the viewer sees
A long 92 ms bar is compared with a short 2 ms bar.

What you need to explain - and why
A cache is a temporary saved answer. Reusing it removes repeated directory work; 92 / 2 = 46 times faster for this lookup alone.

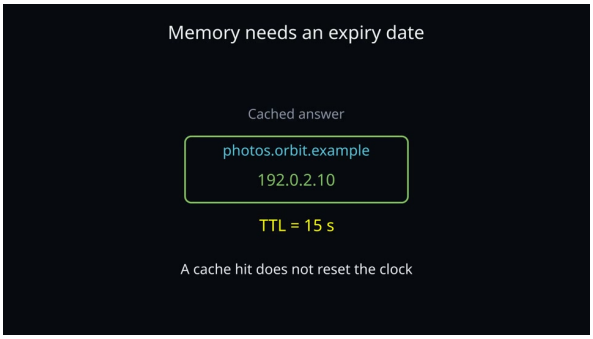
Tone and emphasis
Emphasize the work that disappeared before giving the ratio.

Next: Memory needs an expiry date

Open clip 10 | My phrasing: _____

CLIP 11 / Memory needs an expiry date

05:53.4 - 06:29.1 in full video | 35.7s clip



What the viewer sees
A saved answer has a shrinking timer.

What you need to explain - and why
TTL is the answer's allowed fresh lifetime. Using the saved copy does not restart its clock.

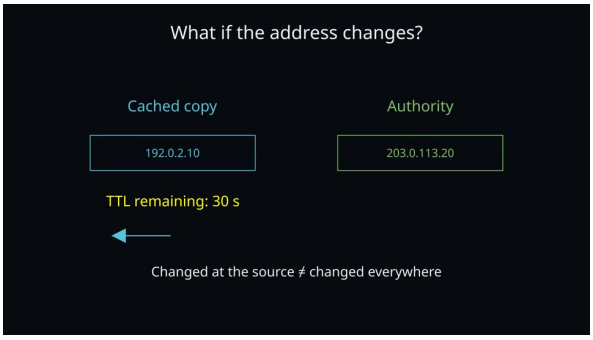
Tone and emphasis
Compare it to an expiry date printed on a note.

Next: What if the address changes?

Open clip 11 | My phrasing: _____

CLIP 12 / What if the address changes?

06:29.1 - 07:04.8 in full video | 35.7s clip



What the viewer sees
The official record changes while the saved copy stays old.

What you need to explain - and why
A source update is not instantly pushed into every cache. The old copy may still be used during its remaining permitted lifetime.

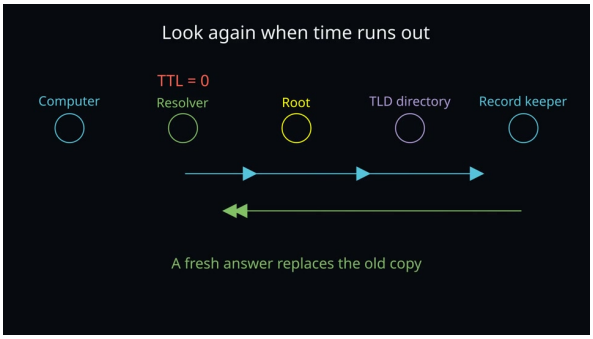
Tone and emphasis
Invite a prediction before revealing the two different addresses.

Next: Look again when time runs out

Open clip 12 | My phrasing: _____

CLIP 13 / Look again when time runs out

07:04.8 - 07:41.3 in full video | 36.5s clip



What the viewer sees
The timer expires and the helper looks again.

What you need to explain - and why
In this normal example, the next request refreshes the expired answer. Real caches may retain some directions and shorten the search.

Tone and emphasis
Connect checking again to the need to notice changes.

Next: Connected, but missing directions

Open clip 13 | My phrasing: _____

CLIP 14 / Connected, but missing directions

07:41.3 - 08:17.8 in full video | 36.5s clip



What the viewer sees
One naming request fails while a known destination remains reachable.

What you need to explain - and why
An app can fail because it cannot find an address even while other traffic continues. This experiment breaks only naming.

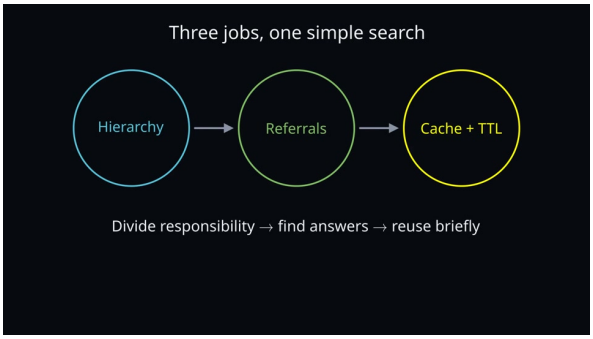
Tone and emphasis
Return to the opening question without presenting DNS as the cause of every outage.

Next: Three jobs, one simple search

Open clip 14 | My phrasing: _____

CLIP 15 / Three jobs, one simple search

08:17.8 - 08:53.9 in full video | 36.1s clip



What the viewer sees
Three circles connect directories, directions and saved answers.

What you need to explain - and why
Hierarchy divides responsibility, referrals guide a search, and caching saves repeated work. Expiry lets old information be replaced.

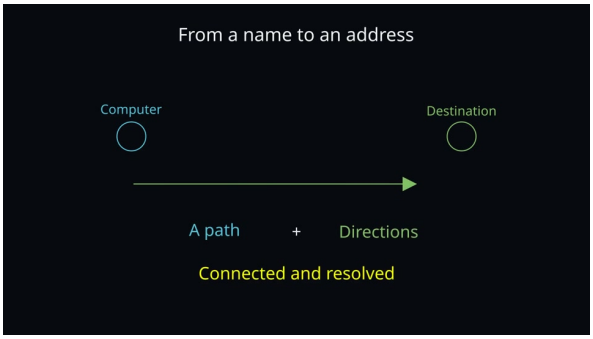
Tone and emphasis
Explain each job with a verb instead of reciting three definitions.

Next: From a name to an address

Open clip 15 | My phrasing: _____

CLIP 16 / From a name to an address

08:53.9 - 09:29.6 in full video | 35.7s clip



What the viewer sees
A computer has both a path and useful directions.

What you need to explain - and why
DNS turns a readable name into contact information through helpers and directories. Saved answers often make that search very short.

Tone and emphasis
Finish on the idea that the website name began as a question.

Close: return to the opening question and its answer.

Open clip 16 | My phrasing: _____